**POLYTECHNIQUE MONTRÉAL**
affiliée à l'Université de Montréal

# A lightweight canonicalization engine in C

**WILLIAM ZHANG**
Département de Electrical Engineering

Mémoire présenté en vue de l'obtention du diplôme de *Maîtrise ès sciences appliquées*
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**A lightweight canonicalization engine in C**

présenté par **William ZHANG**
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a été dûment accepté par le jury d'examen constitué de :

**Danielle DUBOIS**, présidente
**Sébastien VIGOT**, membre et directeur de recherche
**Marie COUTURE**, membre et codirectrice de recherche
**Jean TREMBLAY**, membre
**Joseph BROWN**, membre externe

## DEDICATION

*To all the past, present and future CVXPY contributors*

# ACKNOWLEDGEMENTS

Texte / Text.

# RÉSUMÉ

Le résumé est un bref exposé du sujet traité, des objectifs visés, des hypothèses émises, des méthodes expérimentales utilisées et de l'analyse des résultats obtenus. On y présente également les principales conclusions de la recherche ainsi que ses applications éventuelles. En général, un résumé ne dépasse pas trois pages.

Le résumé doit donner une idée exacte du contenu du mémoire ou de la thèse. Ce ne peut pas être une simple énumération des parties du manuscrit. Le but est de présenter de façon précise et concise la nature, l'envergure de la recherche, les sujets traités, les questions de recherche ou les hypothèses soulevées, les méthodes utilisées, les principaux résultats ainsi que les conclusions retenues. Un résumé ne doit jamais comporter de références ou de figures.

# ABSTRACT

Written in English, the abstract is a brief summary similar to the previous section (Résumé). However, this section is not a word for word translation of the abstract in French.

The abstract is a brief statement of the subject matter, objectives, research questions or hypotheses, experimental methods and analysis of results. It also presents the main research conclusions and their possible applications. In general, an abstract should not exceed three pages.

The abstract should provide an exact idea of the thesis or dissertation's contents and it cannot be a simple enumeration of the manuscript's parts. The goal is to precisely and concisely present the nature and scope of the research. An abstract should never include references or figures. If the thesis or the dissertation is in English, the résumé (French-language abstract) should come first followed by the abstract.

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## LIST OF SYMBOLS AND ACRONYMS

| | |
|---|---|
| AD | Algorithmic Differentiation |
| AML | Algebraic Modeling Language |
| ASA | Affine-Solver-Affine |
| CSC | Compressed Sparse Column |
| CSR | Compressed Sparse Row |
| DAG | Directed Acyclic Graph |
| DCP | Disciplined Convex Programming |
| DPP | Disciplined Parameterized Programming |
| LP | Linear Program |
| MOI | MathOptInterface |
| MPS | Mathematical Programming System |
| NLP | Nonlinear Program |
| QP | Quadratic Program |
| SOCP | Second-Order Cone Program |
| SDP | Semidefinite Program |

# CHAPTER 1 INTRODUCTION

Domain specific modeling languages for convex optimization such as CVXPY [1] and YALMIP [2] allow users to formulate optimization problems in a natural mathematical syntax. A convex optimization problem parametrized by $\theta \in \mathbf{R}^p$ can be written as

$$\begin{array}{ll} \text{minimize} & f_0(x;\theta) \\ \text{subject to} & f_i(x;\theta) \leq 0, \quad i = 1,\ldots,m_1 \\ & g_j(x;\theta) = 0, \quad j = 1,\ldots,m_2 \end{array} \tag{1.1}$$

where $x \in \mathbf{R}^n$ is the optimization variable, the objective $f_0$ and inequality constraints $f_1,\ldots,f_{m_1}$ are convex and the equality constraints $g_1,\ldots,g_{m_2}$ are affine [18]. The parameter $\theta$ specifies data that can change but is fixed and given when solving an instance of the problem. We call the parametrized problem (1.1) a problem *family*; for a fixed value of $\theta$, we call it a problem *instance*.

Through a process called *canonicalization*, these high-level problem descriptions are converted to a *standard conic form* used by many general-purpose convex solvers

$$\begin{array}{ll} \text{minimize} & \frac{1}{2}x^T P(\theta)x + c(\theta)^T x \\ \text{subject to} & b(\theta) - A(\theta)x \in K \end{array} \tag{1.2}$$

where the *problem data* are the symmetric positive semidefinite matrix $P(\theta) \in \mathbf{S}^n_+$, the matrix $A(\theta) \in \mathbf{R}^{m\times n}$, the vectors $b(\theta) \in \mathbf{R}^m$ and $c(\theta) \in \mathbf{R}^n$ and the non-empty closed convex cone $K \subseteq \mathbf{R}^m$. Almost all of optimization practice can be divided into two groups, *modeling* and *solving*. The modeler's role is to formulate problems based on real-world business or engineering applications. The solver's role is to find an optimal solution given the data of a problem instance. Modeling languages serve as the communication bridge between these two groups. When improvements in solvers are made, modeling languages must adapt to allow users to benefit from them. Conversely, it is through natural formulations that modelers can express their requirements and help motivate solver improvements.

While the time taken to solve an optimization problem is often dominated by the solver, the time taken to *canonicalize* (or *compile*) a problem can also be significant, especially for large problems, or when many instances of a family are solved. In the case of CVXPY, this compilation cost lies largely in *matrix stuffing*, the final step of canonicalization where symbolic expression trees are converted to the explicit numerical problem data $P$, $A$, $b$ and $c$.

**Expression trees.** Modeling languages for convex optimization are built on top of a library of functions which can be used to describe problems. These functions are known as *atoms* in CVXPY since they are the fundamental building blocks of the modeling language [3]. An expression is defined recursively as an atom evaluated at *subexpressions*, each of which is itself another expression. Such an expression is naturally stored as an *expression tree*, with atoms at its internal nodes and variables, parameters, and constants at its leaves. After canonicalization the constraint expressions are affine and the objective is quadratic.

**Matrix stuffing in CVXPY.** CVXPY currently canonicalizes a parametrized problem in one of two ways: re-canonicalizing it from scratch for each new value of the parameters, or forming a *parameter map*, an affine map from the parameters to the problem data computed once and re-applied on each solve [26]. We cover CVXPY's matrix stuffing in detail in §2.2. For fast canonicalization, `cvxpygen`, a code-generation extension to CVXPY, can generate a custom C solver for a given problem family, suitable for real-time and embedded optimization [5].

**Matrix stuffing as a derivative.** After canonicalization, $A$ is the Jacobian of the constraints and $P$ the Hessian of the objective, so matrix stuffing amounts to computing a sparse Jacobian and Hessian. Classical methods compute the matrix with several passes of automatic differentiation, using graph coloring to group columns that can be recovered in a single pass [60,62]. Recent vectorised formulations of AD have appeared in specific domains such as statistics [56] and optimal control [55], though not yet for matrix stuffing. We review these methods of computing sparse derivatives in §2.3.

**Other modeling languages.** Other modeling languages produce this data in different ways, a task we call *coefficient extraction*. JuMP and AMPL extract the coefficient data

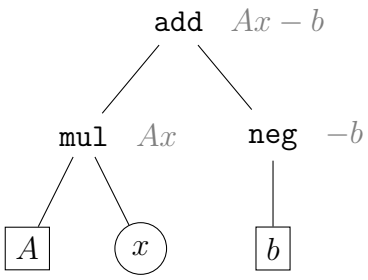



Figure 1.1 Example of an affine expression tree for $Ax - b$. Variables are drawn as circles and constants (or parameters) as boxes.

directly from the algebraic model, with AMPL also recognizing second-order cone structure and rewriting it into the solver's conic form [10, 16, 33]. Through its quadratic-programming interface, CasADi instead forms the Jacobian and Hessian by automatic differentiation, recovering their sparsity pattern by graph coloring [17]. CasADi compiles this into a reusable function, fixing the sparsity pattern once and recomputing only the nonzero values on each solve. We review these methods in much greater detail in §2.4.

**Contributions.** We make the following contributions:

1. We build `SparseDiffengine`, a C library with Python bindings that computes sparse Jacobians and Hessians of vectorized expressions, without graph coloring.
2. We implement a canonicalization backend for CVXPY that builds on `SparseDiffengine`.
3. We benchmark this backend against existing canonicalization approaches, comparing compilation time and memory use across a range of problem families.

All code is open source and available at

`https://github.com/cvxpy/cvxpy`
`https://github.com/SparseDifferentiation/SparseDiffengine`

**Thesis outline.** The rest of this thesis begins with a review of the literature on modeling languages, canonicalization, and computing derivatives in §2. In §3 we present `SparseDiffengine` and the affine and quadratic atoms it uses to canonicalize a problem into its sparse Jacobian and Hessian. In §4 we present benchmark results comparing our method to existing approaches. Finally, in §5 we summarize our contributions and discuss future improvements to the canonicalization backend.

# CHAPTER 2 LITERATURE REVIEW

**History and background** Since the early days of linear programming (LP), practitioners have needed to translate mathematical formulations into data that solvers can process. At first, this meant encoding each coefficient of the problem matrices by hand, a process that is both tedious and error-prone [6]. *Matrix generators* were introduced in the late 1960s to automate this task: they compute the coefficients through explicit loops and write them out in solver-readable formats like MPS [7, 8].

Despite matrix generators removing most of the manual labor, a gap remained. Modelers naturally think in terms of sets, indices, and algebraic expressions, while procedural code constructs the coefficient matrices entry by entry. This mismatch between how a model is expressed and how it is implemented made them difficult to verify, modify, and communicate with others [9]. *Algebraic modeling languages* such as AMPL [10], AIMMS [11] and GAMS [12] closed this gap by accepting problem descriptions that closely follow the math. For a more complete overview of the early history of modeling languages refer to [13].

More recently, modeling languages embedded in general-purpose programming languages have gained popularity, such as Pyomo [14,15] in Python, JuMP [16] in Julia, and CasADi [17] in C++. Other domain-specific modeling languages specialized for convex optimization [18] are built on *disciplined convex programming* (DCP), a set of composition rules under which the convexity of a problem can be verified automatically [3, 4]. These DCP-based languages include CVX [19] in MATLAB, CVXR [20] in R, CVXPY [1] in Python, and Convex.jl [21] in Julia.

**Structure of the chapter** The first two sections focus on CVXPY: it is the most actively maintained DCP implementation, and the work of this thesis aims to be integrated into it. §2.1 covers the *frontend* of canonicalization, which rewrites the user's functional form into a functional conic form of purely affine and quadratic expressions. §2.2 covers its *backend*, *matrix stuffing*, which assembles the problem data from this functional conic form.

Because the problem data assembled this way is exactly a sparse Jacobian of the canonicalized form, §2.3 reviews the methods available for computing sparse derivatives. §2.4 then revisits the same task from the side of other modeling systems: it presents the embedded modeling languages JuMP and CasADi as well as the algebraic language AMPL as instances of that machinery, and compares how each one extracts the problem data with the approach taken in CVXPY. Finally, §2.5 covers the sparse matrix formats used to store the problem data,

along with some useful algorithms for manipulating them.

### 2.1 Canonicalization

CVXPY is a DCP-based modeling language embedded in Python [1], which began in 2013 as an implementation of the modeling principles of CVX [19]. Three fundamental design decisions differed from CVX at the beginning. The first is modularity: a few core data structures are shared by separate components for verification, canonicalization, matrix stuffing, and solver interfaces. The second is the expression tree as the central data structure, which is formed automatically from user code through Python's operator overloading. The third is support for parameters: since canonicalization never looks at the values of constants, much of the compilation work can be reused when a problem is solved repeatedly.

These ideas matured with version 1.0, which redesigned canonicalization as a general reduction-based rewriting system [22]. The new design enabled the addition of other disciplined grammars [23–25], and the early parameter support grew into disciplined parameterized programming (DPP), a restricted grammar under which canonicalized programs can be updated efficiently [26].

The open source project has since grown through contributions from many other enthusiastic developers; a recent paper surveys the new canonicalization and modeling features introduced between versions 1.1 and 1.9 [27].

**Functional form.** A user most naturally expresses a CVXPY problem in *functional* form,

$$\begin{array}{lll} \text{minimize} & f_0(x;\theta) & \\ \text{subject to} & f_i(x;\theta) \le g_i(x;\theta), & i = 1,\ldots,m_1, \\ & h_i(x;\theta) = j_i(x;\theta), & i = 1,\ldots,m_2, \end{array} \tag{2.1}$$

where the optimization variables $x_1, \ldots, x_k$ are collected into a single vector $x \in \mathbf{R}^n$, and likewise the parameters $\theta_1, \ldots, \theta_l$ into $\theta \in \mathbf{R}^p$ [18]. As in §1, (2.1) is a problem family when $\theta$ is left symbolic and a problem instance when $\theta$ is fixed to a numerical value; it generalizes the parametric form (1.1) to the convex/concave inequalities and affine equalities CVXPY accepts. Internally, CVXPY represents each function in (2.1) as an expression tree.

**Disciplined convex programming.** When $f_0$ and all $f_i$ are convex, all $g_i$ are concave, and all $h_i$ and $j_i$ are affine, the functional form defines a convex optimization problem. To certify convexity, CVXPY uses the DCP composition rules to classify the curvature of each

function. Once all functions are classified, a problem is *DCP-compliant* if the objective and each constraint have curvatures compatible with a convex problem: a convex objective, convex $\leq$ concave inequalities, and affine equalities.

Otherwise, if a curvature is incompatible (*e.g.* $f_0$ is concave) or cannot be determined, CVXPY raises an error and terminates. Because these rules are only sufficient for convexity, a problem that is genuinely convex but not expressed in a recognized form may still be rejected.

**Composition rules.** The composition $\phi(f_1(x), f_2(x), \ldots, f_k(x))$ of an atom $\phi$ with subexpressions $f_i$ is convex when $\phi$ is convex and, for each argument $i \in \{1, \ldots, k\}$, one of the following holds:

- $\phi$ is non-decreasing in argument $i$ and $f_i$ is convex, or
- $\phi$ is non-increasing in argument $i$ and $f_i$ is concave, or
- $f_i$ is affine.

The same results hold for concave functions by swapping convex and concave. Affine expressions require $\phi$ and all of its arguments to be affine [3].

**Functional conic form.** Once a problem is verified to be DCP-compliant, the frontend of canonicalization rewrites its functional form (2.1) into an equivalent *functional conic form*

$$\begin{array}{ll} \text{minimize} & f_0(x;\theta) \\ \text{subject to} & \ell_i(x;\theta) \in \mathcal{K}_i, \quad i = 1, \ldots, m, \end{array} \tag{2.2}$$

where $f_0$ is affine or quadratic, each $\ell_i$ is affine, and each $\mathcal{K}_i$ is one of the standard convex cones: the zero, nonnegative, second-order, power, exponential, and positive semidefinite cones.

**Reductions.** This rewriting is carried out as a sequence of *reductions*. Each reduction provides an `apply` step that converts a problem into an equivalent one, and a `retrieve` step that recovers a solution of the original problem from a solution of the rewritten one [22].

A first example is `FlipObjective`, which turns a maximization problem into a minimization one by flipping the sign of the objective; its `retrieve` step simply flips the sign back.

A second example, `CvxAttr2Constr`, handles variable attributes, which are properties declared on a variable that restrict its domain. These include nonnegativity (`nonneg`), positive

semidefiniteness (`PSD`), and explicit variable bounds, amongst others. When the target solver supports an attribute natively, CVXPY leaves it for the solver to enforce; otherwise the reduction creates a new variable that carries no attributes and adds explicit constraints that enforce the attribute. Its `retrieve` step then maps the reformulated solution back onto the original attributed variable.

**Conic reformulation.** The main reduction, `Dcp2Cone`, rewrites every nonlinear atom into conic constraints. The first step of the reduction is to rewrite the problem into *Smith form*, where each nonlinear atom is replaced by a new variable coupled with an equality constraint; every atom then acts only on affine arguments [28]. For a convex atom $\phi$, the Smith form gives an auxiliary variable $t$ with the equality constraint $\phi(x) = t$. The second step relaxes this equality to the inequality $\phi(x) \leq t$. The DCP rules allow a convex atom only where its value is minimized or upper-bounded, so the solver always prefers $t$ as small as possible; the relaxation is therefore *lossless*.

The pair $(x, t)$ then lies in the *epigraph* $\{(x, t) : \phi(x) \leq t\}$ of $\phi$, so we often refer to $t$ as an *epigraph variable*. A concave atom relaxes to $\phi(x) \geq t$ instead, giving a *hypograph variable*. The final step of the reduction expresses each epigraph (hypograph) inequality as the membership of an affine function of $(x, t)$ in a convex cone [29].

**Example of reformulations.** A first example is the $\ell_2$ norm atom $\|x\|_2$, whose epigraph $\|x\|_2 \leq t$ is exactly the second-order cone constraint $(t, x) \in \mathcal{K}_{\text{soc}}$. A second example is the nondifferentiable absolute value atom $|x|$, whose epigraph $|x| \leq t$ can be written smoothly as the pair of affine inequalities $-t \leq x \leq t$. A final example is the sum of the $k$ largest entries of $x$, which expands into a small LP: the epigraph sum_largest$(x, k) \leq t$ holds if and only if there exist a scalar $q$ and a vector $u$ such that

$$q\mathbf{1} + u \geq x, \qquad u \geq 0, \qquad kq + \mathbf{1}^\top u \leq t. \tag{2.3}$$

Here $q$ acts as a threshold and each $u_i$ measures how far the entry $x_i$ rises above it, so that the sum of the $k$ largest entries is bounded by $kq + \mathbf{1}^\top u$ [18].

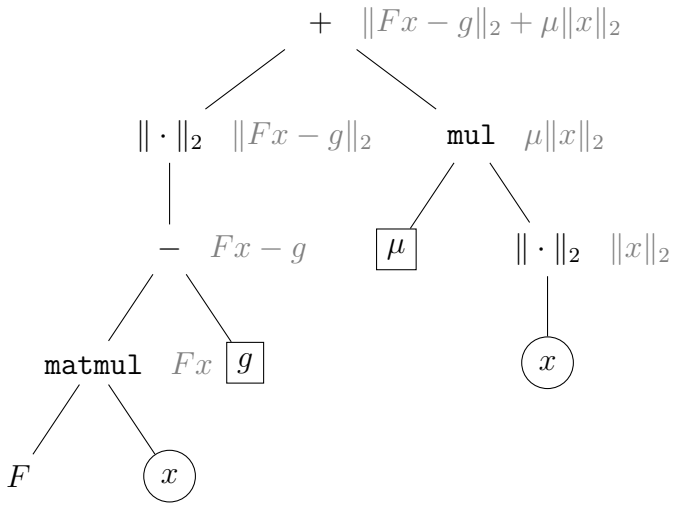
**Example of canonicalization.** To show a concrete example of the frontend of canonicalization, we consider a simple fitting with regularization problem

$$\begin{array}{ll} \text{minimize} & \|Fx - g\|_2 + \mu\|x\|_2 \\ \text{subject to} & x \succeq 0 \end{array} \tag{2.4}$$

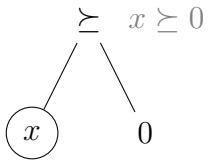
with variable $x \in \mathbf{R}^n$, fixed problem data $F \in \mathbf{R}^{m \times n}$, and parameters $g \in \mathbf{R}^m$ and $\mu \geq 0$. The problem formulation in CVXPY code is

```
import cvxpy as cp
import numpy as np
F = np.random.randn(m, n)
x = cp.Variable(n)
g = cp.Parameter(m)
mu = cp.Parameter(nonneg=True)
objective = cp.Minimize(cp.norm(F @ x - g) + mu * cp.norm(x))
prob = cp.Problem(objective, [x >= 0])
```

where we also use the numerical computation library `NumPy` [30] for the data arrays. The expression tree representation of the problem is shown in 2.1.




(a) Expression tree of the objective



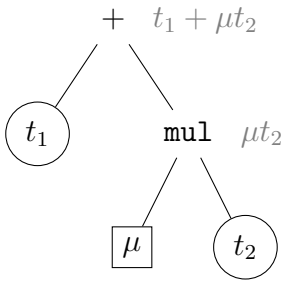

(b) Expression tree of the constraints

Figure 2.1 *Before canonicalization.* CVXPY represents the DCP problem (2.4) as these expression trees. Variables are labeled with a circle, and parameters with a square.

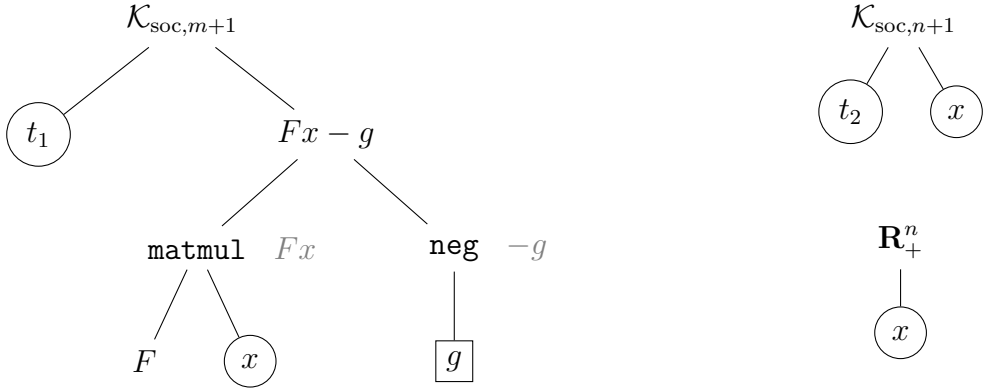
**Conic form expression tree.** After applying the reductions, the problem has been *canonicalized* to

$$\begin{array}{ll} \text{minimize} & t_1 + \mu t_2 \\ \text{subject to} & x \in \mathbf{R}^n_+ \\ & (t_1, Fx - g) \in \mathcal{K}_{\text{soc},m+1} \\ & (t_2, x) \in \mathcal{K}_{\text{soc},n+1} \end{array} \tag{2.5}$$

where $t_1$ and $t_2$ are new epigraph variables, and the cone $\mathcal{K}$ from 1.2 is the product $\mathbf{R}^n_+ \times \mathcal{K}_{\text{soc},m+1} \times \mathcal{K}_{\text{soc},n+1}$. Problem (2.5) is now in the functional conic form introduced above: every constraint is a cone membership, and the objective and constraint expressions are affine. Comparing the DCP trees of 2.1 with the conic-form trees of 2.2 shows the effect of canonicalization: the nonlinear norm atoms have been replaced by epigraph variables, leaving affine expression trees in the variables. The three rows of (2.5) remain three separate cone constraints, each carrying the affine expressions that must lie in its cone; 2.2 accordingly shows one expression tree per constraint, rather than a single combined tree.




(a) Expression tree of the objective



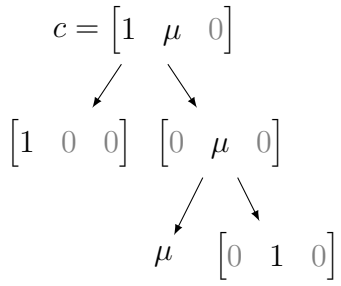

(b) One expression tree per constraint

Figure 2.2 *After canonicalization.* CVXPY represents the conic form (2.5) as these expression trees: the nonlinear norm atoms have become epigraph variables, leaving the objective and the three separate cone constraints affine. Variables are deduplicated and ..

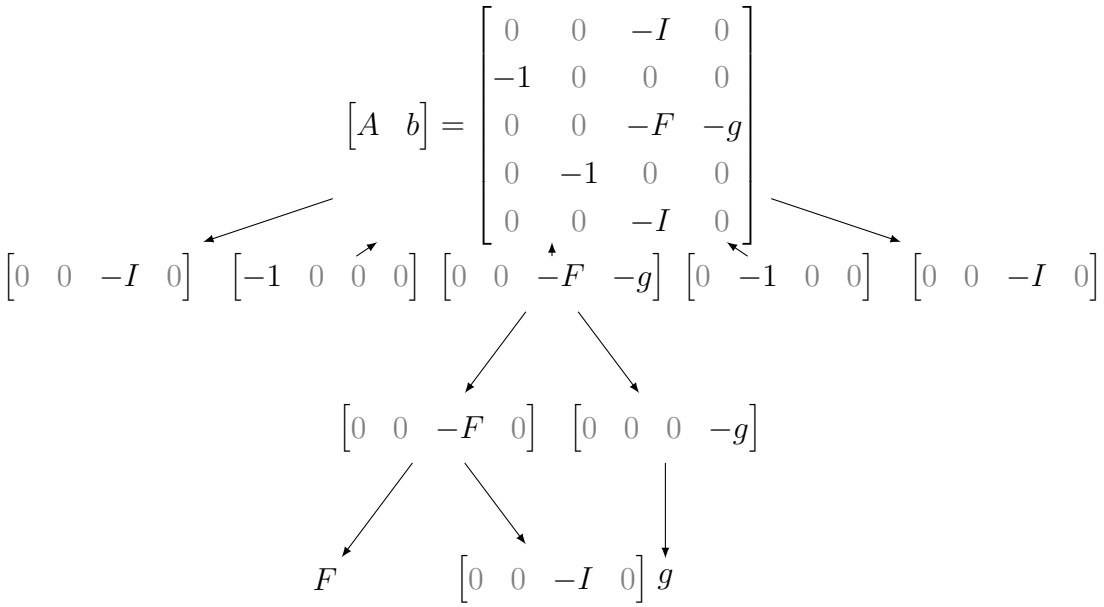
**References.** Michael Grant's papers and PhD thesis on CVX and DCP are the standard references on verifying convexity and canonicalization [3, 4, 29]. The paper by Chu et al. [31] gives another perspective of canonicalization through the *relaxed smith form*, it was used extensively to learn the concept initially. The paper by Agrawal et al. [22] on rewriting systems is a great reference for understanding reductions and the core concepts behind CVXPY. Fi-

nally, the Mosek modeling cookbook is a complementary reference for epigraph reformulations and general modeling best practices [32].

## 2.2 Matrix stuffing




(a) Matrix stuffing of the objective

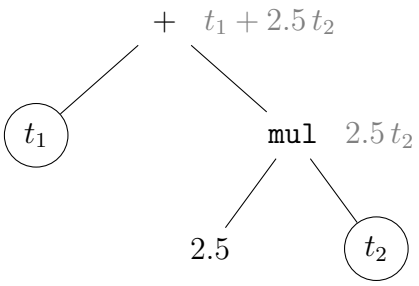



(b) Matrix Stuffing tree

Figure 2.3 Matrix stuffing of problem (2.4) by a recursive walk of the conic-form expression trees of 2.2: each node's matrix is obtained by applying a linear operation to the matrices of its children.

$$Q = \begin{array}{c} \\ t_1 \\ t_2 \\ x \\ d \end{array} \begin{array}{c} \begin{array}{ccc} g^T & \mu & 1 \end{array} \\ \begin{bmatrix} 0 \cdots 0 & 0 & 1 \\ 0 \cdots 0 & 1 & 0 \\ 0 \cdots 0 & 0 & 0 \\ 0 \cdots 0 & 0 & 0 \end{bmatrix} \end{array} \qquad \begin{bmatrix} c \\ d \end{bmatrix} = Q\,\tilde{\theta} = \begin{bmatrix} 1 \\ \mu \\ 0 \\ 0 \end{bmatrix}$$

(a) DPP path: the objective parameter map $Q$ is formed and cached; each re-solve recovers the objective data $(c, d)$ by multiplying $Q$ with the extended parameter vector $\tilde{\theta} = (g, \mu, 1)$.




$$\Rightarrow \quad c = \begin{bmatrix} 1 & 2.5 & 0 \end{bmatrix}$$

(b) `ignore_dpp` path.

Figure 2.4 The two...

**References.**

## 2.3 Computing derivatives

Finding the rate of change, or *derivative*, of a function is a fundamental problem in many fields of science and engineering. Although computing derivatives may seem straightforward at first, there are many approaches, each with its own trade-offs [48]. These approaches differ in four basic choices: whether the function is treated as a *scalar* or *vectorized* expression, whether derivatives are propagated *forward* or in *reverse*, whether the Jacobian or Hessian is assembled *directly* or accessed through an *oracle* of matrix–vector products, and whether the resulting matrix is *sparse* or *dense*.

The problem data for an optimization model is often large and sparse, so we place particular emphasis on computing sparse derivatives in this section. The standard approach in the literature is scalar and oracle-based differentiation [41], where graph coloring is used to recover a sparse matrix in far fewer oracle evaluations. This thesis instead proposes to compute

these sparse derivatives directly on a vectorized graph whose sparsity pattern follows from the structure of the operations, so no coloring is needed.

The remainder of the section is structured as follows: we begin with a brief survey of the standard numerical and symbolic methods, before turning to algorithmic differentiation. Finally, we gather a range of relevant works, organizing them in a table according to their particular approach to differentiation.

**Numerical differentiation.** Other than computing derivatives by hand, the simplest method is to use *numerical differentiation*, which approximates the derivative of a function at a point by evaluating the function at nearby points using a *finite difference* formula similar to either

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}, \tag{2.6}$$

or

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}, \tag{2.7}$$

where $h$ is a small step size. Although this approach is straightforward to implement and can achieve reasonable accuracy, there are important downsides to consider. First, it requires a linear number of function evaluations with respect to the number of input variables, which is expensive for high-dimensional functions. Second, it can easily lead to *truncation errors* (when $h$ is too large) or *rounding errors* (when $h$ is too small) in the computed derivative if the step size is not carefully chosen [42]. Despite these downsides, numerical differentiation is still widely used in practice, especially for verifying the correctness of derivative computations (also known as a *derivative checker*) [43].

**Symbolic differentiation.** Mathematical expressions can be manipulated automatically using simple derivative rules of calculus, such as the product and sum rule. This *symbolic* approach to differentiation is used in computer algebra systems such as Mathematica [44] and Maple [45] as well as the SymPy library in Python [46]. Unlike its numerical counterpart, symbolic differentiation tools compute derivatives as closed-form expressions, which can be simplified and evaluated at any arbitrary point. These closed-form expressions are particularly useful for deriving analytic solutions of optimization problems, and can be further used in mathematical proofs and analyses.

However, a naive implementation of symbolic differentiation can quickly form exponentially large symbolic expressions, which is impractical and inefficient for many problems [41]. Interestingly, when intermediate expressions are stored as *common subexpressions*, and the expression is represented with a *directed acyclic graph* (DAG) instead of a tree, symbolic

differentiation performs the same set of operations as *reverse mode differentiation* [47].

**Algorithmic differentiation.** *Algorithmic differentiation* (AD) treats the function as a *computational graph* of elementary operations, each with a known local derivative rule. The chain rule is then applied to combine these local derivatives throughout the graph, giving the derivative of the whole function. AD is also commonly referred to as *automatic differentiation*; we prefer the term *algorithmic* to emphasize that AD differentiates the algorithm that computes a function, rather than a numerical approximation or a symbolic formula [41]. In this broad sense, AD can refer to any method that applies the chain rule over a computational graph. In the remainder of this section, we introduce the standard scalar and oracle-based differentiation methods, then contrast them with vectorized and direct approaches.

**Scalar differentiation.** The earliest and most foundational form of AD operates on a *scalar* computational graph, where every node is a single elementary operation (an addition, a multiplication, or a univariate function such as sin or exp), and every edge carries the corresponding scalar partial derivative [41, 72]. Since each elementary operation has a simple and known derivative, the chain rule reduces to multiplying and summing scalars along the graph. Early AD tools such as ADOL-C [53] and CppAD [54] form the scalar graph by recording a runtime trace of every elementary operation. Such a trace can be enormous: a single matrix–vector product $Ax$ unrolls into one scalar multiplication for every entry of $A$, together with the additions that sum them, so it quickly reaches millions of nodes for a realistic problem.

**Modes of differentiation.** The chain rule can be applied in different orders, all giving the same result. The two best-known are *forward mode* (or *tangent* mode), which propagates derivatives from the inputs toward the outputs, and *reverse mode* (or *adjoint* mode), which propagates them from the outputs back toward the inputs [48]. The two modes have asymmetric costs: in forward mode the work scales with the number of inputs, and in reverse mode with the number of outputs. Reverse mode is therefore preferred when there are far fewer outputs than inputs, such as computing the gradient of a scalar function. Other mixed-ordering modes exist, commonly grouped under the umbrella term *cross-country* elimination [41]. However, finding the best ordering on a general graph is difficult, so these modes are rarely used in practice [58].

**First- and second-order derivatives.** For a vector-valued map $F : \mathbf{R}^n \to \mathbf{R}^m$ of an input $x \in \mathbf{R}^n$, the first-order derivative is the *Jacobian* $J \in \mathbf{R}^{m \times n}$ of partial derivatives of

the outputs with respect to the inputs,

$$J_{ij} = \frac{\partial F_i}{\partial x_j}, \qquad i = 1, \ldots, m, \quad j = 1, \ldots, n. \tag{2.8}$$

For a scalar function $f : \mathbf{R}^n \to \mathbf{R}$, the second-order derivative is the *Hessian* $H \in \mathbf{R}^{n \times n}$ of second partial derivatives,

$$H_{ij} = \frac{\partial^2 f}{\partial x_i \, \partial x_j}, \qquad i, j = 1, \ldots, n. \tag{2.9}$$

The Hessian is the Jacobian of the gradient of $f$, and is symmetric whenever $f$ is twice continuously differentiable. It is therefore obtained by differentiating the gradient, with no method beyond those used for the Jacobian.

**Oracle-based differentiation.** Instead of $J$ itself, AD returns its action on a vector: the Jacobian-vector product (JVP) $Jv$ in forward mode, and the vector-Jacobian product (VJP) $J^\top w$ in reverse mode (for the Hessian, the analogous Hessian-vector product $Hv$ comes from one forward pass over one reverse pass) [48]. These matrix-vector products are obtained without ever forming $J$, each at about the cost of one function evaluation. Borrowing terminology from numerical linear algebra, we call this the *matrix-free*, or *oracle*, view of AD. The oracle is sufficient for many applications; they never form the matrix and use only its action on vectors, as in backpropagation for training neural networks or Krylov methods for linear systems.

**Recovering the Jacobian.** Recovering $J$ explicitly requires *probing* the oracle with a sequence of seed vectors: the $n$ unit seeds recover it column by column in forward mode, and the $m$ unit seeds recover it row by row in reverse mode (the Hessian is recovered the same way, though its symmetry permits fewer seeds) [41]. When the Jacobian has many rows or columns this becomes prohibitively expensive, and reducing the number of probes is exactly the purpose of the graph-coloring methods described next. Crucially, coloring operates on the oracle alone, independently of how the underlying graph is built, so even fully vectorized frameworks rely on it when they need an assembled sparse Jacobian.

**Graph coloring.** The cost of probing drops when several seeds share a single pass. Two columns of $J$ with no nonzero in a common row are *structurally orthogonal*, so one seed recovers both without their nonzeros colliding. Finding the fewest such groups is a graph-coloring problem on the column intersection graph, whose edges join columns that share a

row [60–62]. The Hessian compresses further: its symmetry admits dedicated star and acyclic colorings that use fewer colors than the general Jacobian case [63].

**Sparse derivatives.** An oracle-based sparse Jacobian is obtained in three stages: *detect* the sparsity pattern, *compress* it by coloring, and *propagate* the compressed seeds through the oracle to recover the nonzero entries [64]. The pattern is computed once and reused, so its cost is amortized across repeated evaluations of the same function. This pipeline underlies sparse AD in oracle-based tools such as CasADi [17], which we revisit in §2.4.

**Vectorized differentiation.** More recent frameworks coarsen the graph so that each node is a whole vector or matrix operation rather than a single scalar. The local derivative of such a node is then a matrix, a block of the Jacobian, and these blocks are combined by the rules of matrix calculus [52]. The resulting graph has far fewer nodes than the scalar trace, which lets frameworks such as JAX [49] differentiate large programs efficiently. Vectorization alone, however, does not change the assembly: these frameworks still expose only the oracle, and still rely on coloring when they need an assembled sparse Jacobian.

**Direct differentiation.** Direct methods go a step further and abandon the oracle altogether. Rather than returning products, they propagate the local Jacobian blocks themselves through the graph and combine them to form $J$ directly, so neither probing nor coloring is needed. Vectorization is the enabler: it gives each block algebraic structure, whereas a scalar node carries only a 0/1 dependency. Recent works apply this idea in statistics [56] and optimal control [55]. Direct propagation wins when the local blocks are well structured and intermediate fill-in stays bounded, while coloring through the oracle can still be cheaper when the intermediates densify. Which assembly is preferable ultimately follows the solver that consumes the result: the conic solvers targeted in this thesis require the matrix explicitly as data, which is exactly what the direct approach produces.

**Overview of recent works.**

**References.** The book by Griewank and Walther [41] is a comprehensive reference on the theory and practice of scalar automatic differentiation, including the graph-elimination view of Jacobian accumulation. The book by Naumann [72] is another great reference that focuses on the implementation of AD in software, with a particular emphasis on the use of operator overloading and source code transformation techniques. The survey by Baydin et al. [48] is

Table 2.1 Differentiation methods classified by the four choices of this section: the *expression type* (scalar or vectorized), the *mode* (forward, reverse, or both), the *assembly* of the Jacobian (formed *directly* by composing local Jacobians, or matrix-free through an *oracle* that returns only matrix–vector products and is reconstructed by seeded sweeps), and the *data format* (sparse or dense). The vectorized, forward, direct, sparse methods (lower block) need no graph coloring, unlike the matrix-free methods above them.

| **Method / work** | **Expression type** | **Mode** | **Assembly** | **Data format** |
|---|---|---|---|---|
| ADOL-C, CppAD [53, 54] | scalar | both | oracle | dense |
| Tapenade, Enzyme [68, 69] | scalar | both | oracle | dense |
| CasADi [17] | scalar / vectorized | both | oracle | sparse |
| JAX [49] | vectorized | both | oracle | dense |
| PyTorch [50] | vectorized | reverse | oracle | dense |
| `torch-sla` [70] | vectorized | reverse | oracle | sparse |
| CPR, Coleman–Moré [60, 61] | scalar | forward | oracle | sparse |
| Hill–Dalle [64] | scalar | both | oracle | sparse |
| Vector forward mode [71] | scalar | forward | oracle | dense |
| Zou and Jiang [55] | vectorized | forward | direct | sparse |
| Kwok et al. [56] | vectorized | forward | direct | sparse |
| $\nabla$SD [57] | vectorized | forward | direct | sparse |
| **This thesis** | **vectorized** | **forward** | **direct** | **sparse** |

an excellent and relatively up-to-date overview of the various methods for automatic differentiation in machine learning. For the combinatorial side of sparse derivative computation, the survey of Gebremedhin et al. [62] is the standard entry point, and the illustrated guide of Hill and Dalle [73] gives an accessible modern tutorial on the detection–coloring–compression pipeline.

## 2.4 Coefficient extraction

**`JuMP`.** JuMP is an algebraic modeling language embedded in Julia whose development started in 2012 at the MIT Operations Research Center by Iain Dunning, Joey Huchette, and Miles Lubin, with a first public release in 2013 [16,33]. It is now maintained by a team of core contributors, with Oscar Dowson as its most active maintainer. Unlike CVXPY, JuMP performs no convexity verification; it belongs to the derivative-based modeling paradigm, where the user is responsible for the properties of the formulation. Coefficient extraction in JuMP is *eager*: the `@constraint` and `@objective` macros rewrite the user's code at parse time into loops that build each affine expression as a sparse list of terms (an ordered dictionary mapping variables to coefficients). This macro rewriting avoids the asymptotic blowup of naive operator overloading, where summing $d$ terms with repeated calls to `+` copies partial sums of growing size. As a consequence, the problem data is scalarized one constraint at a

time while the model is being built, whereas CVXPY retains whole expression trees until solve time.

**`MathOptInterface`.** JuMP communicates with solvers through MathOptInterface (MOI), an abstraction layer whose standard form is to minimize $f_0(x)$ subject to constraints $f_i(x) \in S_i$, where the functions $f_i$ and sets $S_i$ are drawn from predefined lists [34]. There is deliberately no single canonical form: each solver advertises the function-set pairs it supports natively, and unsupported constraints are rewritten by *bridges*, local transformations selected per constraint type by a shortest-path search over a hypergraph of possible reformulations. Bridges play the same role as CVXPY's reductions, but they are chosen just-in-time as constraints are added to the model, rather than applied as a global pass over the whole problem. The constraint matrix only materializes inside the solver wrapper. When the model is copied to the solver, MOI assembles the affine terms of all constraints into a single sparse matrix in compressed sparse column (CSC) format through a two-pass, sort-free procedure: a first pass counts the nonzeros falling in each column and a prefix sum turns these counts into the column pointers, sizing the row and value arrays to exactly the number of nonzeros; a second pass scatters each term into its slot using a per-column cursor. The column indices are stored either one-based, for an allocation-free view as a native sparse matrix, or zero-based, so the arrays can be handed directly to a C solver. This assembly is the same coordinate-to-CSC computation of §2.5, but it is carried out for a single fully numeric instance: the standard form $f_i(x) \in S_i$ admits no symbolic parameters, so re-solving with changed data requires rebuilding the affine terms or modifying individual coefficients in place and re-copying to the solver. There is no precompiled map from a parameter vector to the problem data of the kind CVXPY caches (**??**); even `ParametricOptInterface.jl`, which adds parameters to the JuMP ecosystem, substitutes their fixed values and re-optimizes rather than applying a reusable sparse parameter-to-data map.

**`AMPL`.**

**`CasADi`.**

**References.** The MathOptInterface paper [34] is the best reference on the design of JuMP's solver abstraction layer and its bridge system. David Gay's technical reports [35, 36] remain the authoritative description of the `.nl` format and the AMPL solver interface. Finally, the CasADi paper [17] is a self-contained reference covering its expression graphs, AD implementation, and solver interfaces.

Table 2.2 Comparison of problem-data extraction methods across modeling languages.

| | **JuMP** | **AMPL** | **CasADi** | **CVXPY** |
|---|---|---|---|---|
| Front-end paradigm | Algebraic modeling | Algebraic, file-based | Symbolic graphs + AD | Disciplined convex (DCP) |
| Problem classes | LP, MILP, QP, conic, NLP | LP, MILP, NLP (conic via recent drivers) | NLP, QP / conic (QP-shaped) | Convex: LP, QP, SOCP, SDP, EXP |
| Path to conic solvers | Native cones + bridges (MOI) | Solver-side, recent drivers | `soc` helper on the QP path | Automatic DCP reduction |
| Problem-data mechanism | Eager, per scalar constraint | `.nl` file; nonlinear via solver AD | Explicit $H, A$ by AD on the graph | Affine-tree stuffing + parameter map |

## 2.5 Sparse matrices

Matrices that contain many number of zero entries are known as *sparse matrices*. The simplest sparse matrix data structure is the *coordinate* or *triplet* format. It consists of two integer index arrays `rows` and `cols` and a floating point value array `vals` of length equal to the number of *nonzero entries* in the matrix.

For example, the matrix

$$\begin{bmatrix} 0 & 0 & 3.2 & 0 \\ 4.1 & 1.8 & 0 & 2.2 \\ 0 & 2.8 & 0 & 3.4 \end{bmatrix} \tag{2.10}$$

can be represented in coordinate format as

```
int    rows[] = {0,  1,   1,   1,   2,   2};
int    cols[] = {2,  0,   1,   3,   1,   3};
double vals[] = {3.2, 4.1, 1.8, 2.2, 2.8, 3.4};
```

where we use zero-based indexing and array declaration notation from the C programming language [74]. The coordinate format is the easiest to understand and form, but it is not effective for most operations on sparse matrices; the *compressed sparse row* or CSR format is commonly used instead.

**Compressed sparse row format.** The CSR format differs from the coordinate format in that it replaces the `rows` array with a row pointer array `idxptr` of length equal to the number of rows plus one. The row pointer array indicates the starting index of each row in the `cols` arrays. For example, the matrix in (2.10) can be represented in CSR format as

```
int idxptr[] = {0,  1,             4,        6};
```

```
int    cols[] = {2,   0,    1,    3,    1,    3};
double vals[] = {3.2, 4.1, 1.8, 2.2, 2.8, 3.4};
```

where we deliberately format `idxptr` to align with the start of each row in the `cols` arrays. The benefits of the CSR format are more than a reduced size of the `idxptr` array compared to the `rows` array. In the CSR format, accessing a row takes constant time and is very *cache-friendly* since the nonzero entries are stored contiguously in memory. For this reason, the CSR format is the default for *iterative methods* to solve sparse linear systems [75].

**Compressed sparse column format.** However, one downside of the CSR format is that accessing a column is much more difficult; one would need to look at every row to find the nonzero entries in a column. An alternative format is the *compressed sparse column* or CSC format, which swaps the role of the `idxptr` array to refer to column indices instead. In contrast to iterative methods, the CSC format is the preferred format for *direct methods* to solve sparse linear systems [76].

**Transpose.**

**Which format to use?**

**Matrix-vector products.** Mention that both CSR and CSC can do efficient mat-vec, but the algorithm must adapt to the underlying data structure.

**Matrix-matrx products.** Mention SpGEMM. Mention Gustavson's algorithm and the standard one CSR and CSC method. Discuss the tradeoffs.

**Other formats.** The coordinate, CSR and CSC formats are the most widely used sparse matrix formats, but many others exist for different specialized use cases. The scientific Python sparse library `scipy.sparse` [77] allows users to form other types of sparse matrices such as the list of lists (LIL) matrix, the dictionary of keys (DOK) matrix, and the block-sparse row (BSR) matrix.

**Sparse matrix struct in C.** A CSR matrix struct in C looks like this.

```
typedef struct CSR_Matrix {
    int *p;      /* row pointers, size m+1 */
    int *i;      /* column indices, size nnz */
```

```
    double *x;   /* values, size nnz       */
    int m, n, nnz;
} CSR_Matrix;
```

**References.** Although the work in this thesis does not necessarily revolve around solving sparse linear systems, a lot of techniques for manipulating sparse matrices we use were inspired by the literature. In particular, the book by Davis [76] was used extensively throughout the development of our methods. The book by Duff [78] is another great reference to learn more about the history and development of sparse matrix algorithms. Another reference is the review article by the "gang of four" (Gill, Murray, Saunders and Wright) from the Systems Optimization Laboratory at Stanford University [79] which provides a high-level overview of various applications of sparse matrix methods in optimization.

# CHAPTER 3 A lightweight canonicalization engine

This chapter presents a minimal implementation of a subset of affine atoms in the `SparseDiffEngine` package, and shows how they compute the sparse constant Jacobian of a set of affine expressions. We start from a *canonicalized* problem in standard form, produced by the canonicalization of §2.1. The goal is the same as in §2.2: form the problem data. §3.2 describes the data structures, §3.3 the atom implementations, and §3.6 the problem class and its CVXPY interface. We assume the sparse-matrix background of §2.5.

## 3.1 Keeping parameters, splitting the walk

Matrix stuffing already walked the objective tree of our running example. Figure 2.3 annotated each node of the canonicalized objective $t_1 + \mu t_2$ of (2.5) with the row of $c$ it produces, ending in $c = \begin{bmatrix} 1 & \mu & 0 \end{bmatrix}$ at the root. The engine walks the very same tree of 2.2, but it differs from matrix stuffing in two ways.

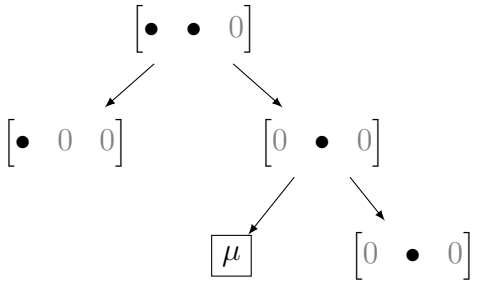
First, it keeps the parameter. The `ignore_dpp` path of §2.2 begins with `EvalParams`, which deletes the parameter leaf $\mu$ and replaces it by a constant holding its current value (2.4b); the walk that follows is parameter-free and must be redone from scratch whenever $\mu$ changes. We instead leave $\mu$ in place as a leaf node, so the tree that produces $c$ is fixed once and reused across solves.

Second, it splits the single walk into two passes over that fixed tree, shown in 3.1. A *structural* pass fixes the sparsity pattern of every node's Jacobian from the patterns of its children, independent of the value of $\mu$ (3.1a). A *value* pass then fills the numbers into that fixed pattern, reading $\mu$'s current value at the leaf (3.1b). The structural pass runs once per problem family; only the value pass repeats on each solve, whereas matrix stuffing fuses the two and recomputes both on every change of $\mu$. These passes are the `jacobian_init` and `eval_jacobian` functions that the rest of the chapter develops atom by atom.

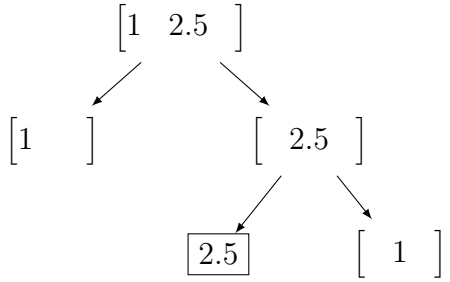
On this objective the saving is admittedly hard to see: the structural pass over $t_1 + \mu t_2$ is trivial, and one could fairly object that re-evaluating so small a tree from scratch costs nothing. The split earns its keep where the structural work is both expensive and shared across solves. The clearest case in our running example is the constraint tree of 2.3, which contains the product $Fx$ (3.2). The Jacobian of $Fx$ is a sparse matrix–matrix product handled by the matrix multiplication atom of §3.3, and its $-F$ block has a sparsity *pattern* that is the costly part to derive and is independent of any parameter value. Because the parameter

$g$ enters the same constraint, the `ignore_dpp` path re-derives this block from scratch each time $g$ changes, while the engine fixes its pattern once in the structural pass (3.2a) and only refills values (3.2b). The per-solve cost this removes, negligible for the scalar objective but substantial for problems with large sparse products solved repeatedly, is quantified in §4.

Like `cvxcore`, which mirrors the Python expression trees into equivalent C++ objects, the engine mirrors each CVXPY atom onto a C counterpart and rebuilds the canonicalized trees as the C `expr` trees of §3.2; both passes then run entirely in C.




(a) Structural pass: a bullet marks a structural nonzero.



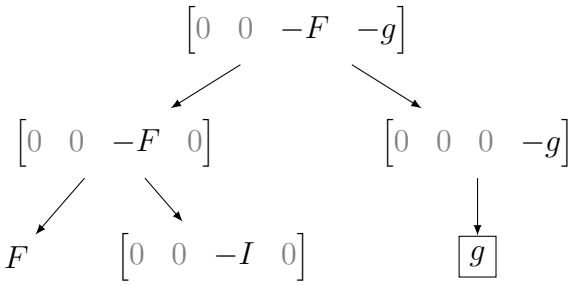

(b) Value pass: entries refilled here for $\mu = 2.5$.

Figure 3.1 The engine's two passes over the objective tree of 2.2, with the parameter leaf $\mu$ kept across both.

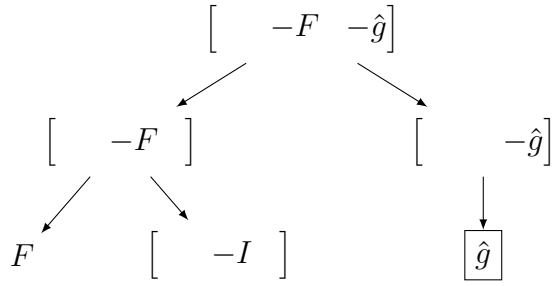
### 3.2 Fundamental data structures

The main data structure in the engine is the expression struct.

```
typedef struct expr {
    int   d1, d2, size; /* shape (d1 x d2) and size = d1*d2 */
    int   n_vars;       /* total number of variables     */
    int   var_id;       /* offset into u, or NOT_A_VARIABLE */
```




(a) Structural pass: the matmul block $-F$ inherits the pattern of $F$ and is independent of $g$.




(b) Value pass: only the values are written into that pattern, here for $g = \hat{g}$. Between solves only the parameter block $-\hat{g}$ changes; the matmul block $-F$ is untouched.

Figure 3.2 The two passes on the $Fx - g$ subtree of the constraint tree of 2.3. The constant $F$ enters through the matmul atom, whose block $-F$ has a sparsity pattern fixed once; across solves only the parameter $g$ changes.

```
    struct expr *left;   /* left child (NULL for leaves)  */
    struct expr *right;  /* right child (NULL for leaves) */

    double    *value;    /* forward-pass output, size entries */
    CSR_Matrix *jacobian; /* size x n_vars sparse Jacobian */

    forward_fn     forward;
    jacobian_init_fn jacobian_init_impl;
    eval_jacobian_fn eval_jacobian;
} expr;
```

Each node carries a `value`, the output of its forward pass, and a sparse Jacobian in CSR form. Each node also implements three functions: `forward`, `jacobian_init`, and `eval_jacobian`.

**Leaf nodes.** The leaves are variables and parameters.

### 3.3 Atom implementations

An atom takes one or two expression operands and produces an expression. Each atom implements the three functions: `forward` computes the value, `jacobian_init` fixes the Jacobian's sparsity pattern once, and `eval_jacobian` fills its values on each solve. We give the full C implementation of one atom, `add`, to establish the pattern, and describe the rest in prose, each anchored by its CVXPY expression. Throughout, `x` and `y` denote variables, `a` and `b` parameters, and `F` a constant matrix, all of compatible shape.

**Addition.** The addition atom sums two expressions elementwise.

```
add_expr = cp.add(x, y)
```

The implementation below is the template every other atom follows. The forward pass evaluates the children and sums their values. `jacobian_init` allocates the output Jacobian, whose number of nonzeros is at most the sum of the children's, and fixes its pattern. `eval_jacobian` sums the children's Jacobians into it.

```
static void forward(expr *node, const double *u)
{   /* forward pass for children */
    node->left->forward(node->left, u);
    node->right->forward(node->right, u);
    for (int i = 0; i < node->size; i++)
    {
        node->value[i] = node->left->value[i] + node->right->value[i];
    }
}

static void jacobian_init_impl(expr *node)
{
    /* initialize children's jacobians */
    jacobian_init(node->left);
    jacobian_init(node->right);

    /* we never have to store more than the sum of children's nnz */
    int nnz_max = node->left->jacobian->nnz + node->right->jacobian->nnz;
    node->jacobian = new_csr_matrix(node->size, node->n_vars, nnz_max);

    /* fill sparsity pattern */
    sum_csr_alloc();
```

```
}

static void eval_jacobian(expr *node)
{
    /* evaluate children's jacobians */
    node->left->eval_jacobian(node->left);
    node->right->eval_jacobian(node->right);

    /* sum children's jacobians */
    sum_csr_fill_values();
}
```

**Negation.** The negation atom negates an expression elementwise. The forward pass flips the sign of each value. The Jacobian has the child's pattern, so `jacobian_init` copies it and `eval_jacobian` negates its values.

**Transpose.** The transpose atom transposes an expression.

```
transpose_expr = cp.transpose(x)
```

The forward pass writes the transpose into the row-major output buffer. Transposing only relabels the output entries, so the Jacobian is a fixed row permutation of the child's: `jacobian_init` records the permutation, and `eval_jacobian` copies the child's values into the permuted rows.

**Horizontal stack.** The horizontal stack atom concatenates expressions of equal row count side by side.

```
hstack_expr = cp.hstack([x, y])
```

The forward pass concatenates the arguments' values. The Jacobian stacks the arguments' Jacobian rows: `jacobian_init` lays out the combined pattern, and `eval_jacobian` copies each argument's values contiguously into place.

**Vertical stack.** The vertical stack atom concatenates expressions of equal column count.

```
vstack_expr = cp.vstack([x, y])
```

It needs no new code. The engine uses the identity `vstack([A, B]) = transpose(hstack([transpose(A) transpose(B)]))`, inheriting the forward and Jacobian behaviour of transpose and hstack.

**Indexing.** The indexing atom selects a subset of an expression's entries.

```
index_expr = x[0:3]
```

The forward pass gathers the selected entries. The Jacobian is the corresponding subset of the child's rows: `jacobian_init` records which rows are selected, and `eval_jacobian` copies their values.

**Sum.** The sum atom reduces all entries of an expression to a scalar.

```
sum_expr = cp.sum(x)
```

The forward pass accumulates the child's values into a scalar. The Jacobian is the column-wise sum of the child's Jacobian, a single row: `jacobian_init` fixes its pattern, and `eval_jacobian` accumulates the child's values into it.

**Elementwise multiplication.** The elementwise multiplication atom scales an expression entrywise by data `a`, constant or parameter.

```
elemwise_mult_expr = cp.multiply(x, a)
```

The forward pass multiplies the child's values by `a`, refreshing `a` first if it is a parameter. Scaling row $i$ by $a_i$ leaves the child's pattern unchanged, so `jacobian_init` copies it and `eval_jacobian` scales each entry of row $i$ by $a_i$. A change in `a` is absorbed entirely at evaluation.

**Matrix multiplication.** The matrix multiplication atom takes a constant matrix $A \in \mathbf{R}^{m\times k}$ and an expression $x$ with $x \in \mathbf{R}^k$, and produces $Ax \in \mathbf{R}^m$. The canonicalized problem is affine, so one operand is always data; we take it to be the left operand $A$, with the variable-bearing expression as the right child.

```
matmul_expr = A @ x
```

The forward pass computes the sparse matrix–vector product $y = Ax$. If $x$ has Jacobian $J_x \in \mathbf{R}^{k\times n_{\text{vars}}}$, then $y$ has Jacobian $J_y = AJ_x$, a sparse matrix–matrix product. Its pattern depends only on the patterns of $A$ and $J_x$.

This is the one atom where the two-phase split does real work. We factor the product into a *symbolic* routine, which allocates $J_y$ and fills its pattern from the patterns of $A$ and $J_x$, and a *numeric* routine, which fills the values into that pattern. `jacobian_init` runs the symbolic routine once; `eval_jacobian` runs only the numeric routine on each solve. The symbolic

step is the expensive part of a sparse matrix–matrix product, and it is hoisted out of the per-solve path.

### 3.4 Problem class

The atoms compute the Jacobian of a single expression. A `Problem` ties together the full set of constraint expression trees and assembles their root Jacobians into the global problem data $[A\ b]$ and objective data $[c\ d]$.

At construction the problem runs the one-time *initialization* phase. It walks each expression tree once, calling `jacobian_init` bottom-up so every node allocates its Jacobian and fixes its pattern. The pattern of an affine expression is independent of its parameter values, so this walk runs once per problem family. The root Jacobians are concatenated into the global CSR buffer for $A$ using the constraint row offsets of §2.2, fixing the pattern of the problem data as a whole.

### 3.5 Parameter updates

The *evaluation* phase runs on every solve. Given new parameter values $\theta$, the problem refreshes the leaf parameters and calls `eval_jacobian` bottom-up, filling the values of each pre-allocated Jacobian without touching its pattern. The forward pass supplies the constant column $b$, and the root Jacobians are copied into the global buffer. A parameter update therefore costs only a forward pass and a numeric Jacobian evaluation: no allocation, no symbolic work, no graph mutation.

This is the saving over the `ignore_dpp` path of §2.2. That path is general — it places no DPP restriction on how parameters enter — but re-derives the symbolic structure on every change of $\theta$. Our two-phase design keeps that generality while paying the symbolic cost once, recovering the re-solve speed of the cached DPP path without materializing the three-dimensional parameter tensor.

### 3.6 CVXPY interface

We expose the `Problem` class and the atom constructors through a thin set of Python bindings. The engine then acts as a *canonicalization backend*, occupying the place of `cvxcore` and the `SciPy` backend of §2.2: it consumes the canonicalized affine expression trees from the reductions and returns the problem data $(A, b, c, d)$.

CVXPY hands the backend its canonicalized expression trees. The binding mirrors each

CVXPY atom onto its engine counterpart, builds the C expression trees, runs the initialization phase once, and stores the handle on the problem. On each solve with new parameter values, CVXPY calls the evaluation phase to obtain the updated data, which is forwarded to the cone solver. The binding forwards only pointers to pre-allocated buffers, so no problem data is copied between Python and C in the inner loop.

# CHAPTER 4 Benchmark results

In this chapter we evaluate the engine of §3 against CVXPY's existing canonicalization path. The central question is the cost of repeatedly forming the problem data for a parameterized problem family, which is exactly the regime our two-phase design targets.

## 4.1 Benchmark Setup

We benchmark on the public CVXPY benchmark suite [80], 25 real-world problems (linear and quadratic programs, least squares, portfolio models, robust regression, and conic and semidefinite programs) maintained under *airspeed velocity* (`asv`) [81] with seeded, reproducible data. Three problems need the unsupported `kron` atom and are excluded; the remaining 22 run on both backends.

We time *canonicalization* only—the construction of the problem data $(A, b, c, P)$ from the expression tree—and exclude the numerical solve, since that is what the engine changes. Concretely we time `get_problem_data` with the target solver fixed to Clarabel [82]: the `CPP` (`cvxcore`) baseline is the default DPP path, and the C engine is selected with `ignore_dpp`.

```
prob.get_problem_data(solver=cp.CLARABEL)              # CPP baseline (DPP
    tensor)
prob.get_problem_data(solver=cp.CLARABEL, ignore_dpp=True) # our C engine
```

We report two regimes: *cold-start* (a fresh `Problem` built with nothing cached, averaged over 3 builds) and *warm-restart* (parameter values reassigned on a warm cached chain, averaged over 20 re-compiles)—the latter being the regime the engine targets. Alongside these wall-clock times we report *peak memory*, and summarize across the suite with the *geometric mean* of the per-problem engine/`CPP` ratios, the standard scale-free statistic for timings that span orders of magnitude [83].

All measurements run single-threaded on a MacBook Pro (13-inch, M2, 2022): 8-core Apple M2, 24 GB RAM, macOS 26.3, with Python 3.14, CVXPY 1.10 (dev), NumPy 2.4 [30], and SciPy 1.16 [77]. Versions are pinned and the suite and engine are public, so the configuration is reproducible.

### 4.2 Results across the benchmark suite

Table 4.1 reports cold-start compilation across the 22 runnable benchmarks, and Table 4.2 reports warm-restart re-compilation on the parametric subset. In both, the ratio is C engine/CPP, so a value below 1 means the engine is faster; the final row gives the geometric mean of the per-problem ratios.

Table 4.1 Cold compilation time: a fresh problem formed and canonicalized from scratch, C engine versus CVXPY's `CPP` backend (22 runnable benchmarks). Ratio = C engine / `CPP`; values below 1 favor the engine.

| Benchmark | `CPP` (ms) | C engine (ms) | Ratio |
|---|---|---|---|
| Yitzhaki | 845 | 69 | **0.08** |
| Cajas | 2154 | 179 | **0.08** |
| SimpleQP | 2882 | 295 | 0.10 |
| LeastSquares | 2313 | 263 | 0.11 |
| HuberRegression | 2559 | 298 | 0.12 |
| SimpleLP | 5656 | 747 | 0.13 |
| SlowPruning | 1726 | 283 | 0.16 |
| SVMWithL1Regularization | 2593 | 411 | 0.16 |
| TvInpainting | 879 | 164 | 0.19 |
| SimpleScalarParametrizedLP | 1066 | 223 | 0.21 |
| SimpleFullyParametrizedLP | 463 | 110 | 0.24 |
| ParametrizedQP | 2645 | 631 | 0.24 |
| ConeMatrixStuffing | 2589 | 819 | 0.32 |
| ParamSmallMatrixStuffing | 118 | 40 | 0.34 |
| OptimalAdvertising | 2202 | 805 | 0.37 |
| ParamConeMatrixStuffing | 69 | 27 | 0.38 |
| SmallMatrixStuffing | 1633 | 627 | 0.38 |
| CVaR | 7128 | 6224 | 0.87 |
| FactorCovarianceModel | 1157 | 1117 | 0.97 |
| Murray | 1052 | 1627 | 1.55 |
| ConvexPlasticity | 81 | 220 | 2.63 |
| SemidefiniteProgramming | 995 | 8886 | **8.90** |
| **Geometric mean** | | | **0.32** |

**Reading.** In aggregate the engine is about 3× faster on cold builds (geometric-mean ratio 0.32) and near parity on parametric re-compiles (1.18), but the per-problem split is the more informative story. On *cold* compilation the engine wins on the bulk of the LP, QP, portfolio, and conic problems—typically 3–12× faster, since it extracts the problem data directly from

Table 4.2 Parametric re-compilation time on a warm cached chain: parameter values are reassigned and the problem data rebuilt (the engine's target regime). Ratio = C engine / `CPP`.

| Benchmark | #par | `CPP` (ms) | C engine (ms) | Ratio |
|---|---|---|---|---|
| ParamSmallMatrixStuffing | 3 | 2.23 | 0.65 | **0.29** |
| ParamConeMatrixStuffing | 3 | 1.19 | 0.40 | 0.33 |
| SimpleFullyParametrizedLP | 1 | 50.8 | 36.8 | 0.72 |
| ParametrizedQP | 2 | 333 | 309 | 0.93 |
| SimpleScalarParametrizedLP | 1 | 66.2 | 67.9 | 1.03 |
| FactorCovarianceModel | 2 | 21.3 | 72.2 | 3.39 |
| SVMWithL1Regularization | 1 | 40.2 | 150 | 3.72 |
| ConvexPlasticity | 3 | 43.0 | 194 | 4.52 |
| **Geometric mean** | | | | **1.18** |

the expression trees while `CPP` pays the full parameter-tensor build. It loses on `Murray` (1.55×), `ConvexPlasticity` (2.63×), and the `SemidefiniteProgramming` outlier (8.9×), and reaches parity on `FactorCovarianceModel` (0.97×). On *parametric re-compilation* the picture is mixed and use-case dependent: the engine is faster on the matrix-stuffing and fully-parametrized-LP re-solves (0.29–0.72×) and near parity on `ParametrizedQP`, but 3.4–4.5× slower on `FactorCovarianceModel`, `SVM`, and `ConvexPlasticity`, where `CPP`'s cached DPP parameter→data tensor (one sparse matmul per solve) is hard to beat. The largest memory wins come from keeping constants sparse: eliminating an eager $25\,000 \times 25\,000$ dense densification ($\approx 5$ GB) on `FactorCovarianceModel`, and shrinking the `ConvexPlasticity` Hessian block from 9M to 51k nonzeros by routing a mostly-zero `quad_form` matrix to the sparse binding.

### 4.3 Scaling with problem size

The cross-suite tables fix the problem size; here we instead sweep it. We deep-dive three parameterized problems—parametric least squares, a factor-covariance portfolio, and an energy-storage scheduling model—and re-measure cold-start time, warm-restart time, and peak memory as the size grows. All three are genuinely parametric, so the warm-restart curve measures the engine's target regime: refilling values on a fixed sparsity pattern as the parameters change. For each panel we plot `CPP` against the C engine on log–log axes.

### 4.3.1 Parametric least squares

The bread-and-butter case: fit $x$ to noisy observations with the data matrix $A$ and right-hand side $b$ supplied as parameters, so a re-solve only reassigns $A$ and $b$. The size knobs are the number of observations $m$ and variables $n$.

```
import cvxpy as cp

A = cp.Parameter((m, n))    # data matrix  (reassigned each solve)
b = cp.Parameter(m)         # observations (reassigned each solve)

x = cp.Variable(n)
objective  = cp.Minimize(cp.sum_squares(A @ x - b))
constraints = [0 <= x, x <= 1]
prob = cp.Problem(objective, constraints)
```

We sweep $n \in \{250, 500, 1000, 2000, 4000\}$ with $m = 3n$ (overdetermined) and measure both backends in the *ignore-DPP* regime: the canonicalization is rebuilt from scratch whenever the parameters change, which is the engine's intended use case. The C engine carries the parameters natively and re-extracts values on a fixed sparsity pattern; the `CPP` baseline substitutes the parameter values (the `EvalParams` reduction) and re-canonicalizes the resulting constant-data problem. We deliberately avoid CPP's DPP parameter-tensor path here: because the data matrix $A$ is itself an $m \times n$ parameter, that tensor has $\Theta(mn)$ entries and its construction dominates everything else—tens of seconds and gigabytes already at $n = 500$—so it is neither representative nor the comparison of interest. Cold-start and warm-restart times are medians over three repetitions; peak memory is the process resident-set high-water mark (`ru_maxrss`), measured in a fresh subprocess per point so that, unlike Python-level `tracemalloc`, it also captures the engine's C-side allocations.

Cold-start (Figure 4.1) favours the engine by roughly 4× across the range (ratio 0.21–0.27), both backends scaling as $\sim n^{2.1}$. The gap widens on warm-restart (Figure 4.2): `CPP` re-canonicalizes in essentially its cold-start time (9.1 s at $n = 4000$), whereas the engine re-extracts on the cached pattern in 863 ms—a 10–12× speed-up that is precisely the non-DPP re-solve regime the engine targets. Peak memory (Figure 4.3) is comparable: both grow as $\sim n^{1.6}$ and are dominated by the dense $A$, with the engine 2–30% heavier in resident set. Its own internal accounting (the engine's `verbose` "peak memory") grows as $\sim n^{2.0}$, from 14 MB at $n = 250$ to 3.7 GB at $n = 4000$, yet stays below the process RSS throughout.

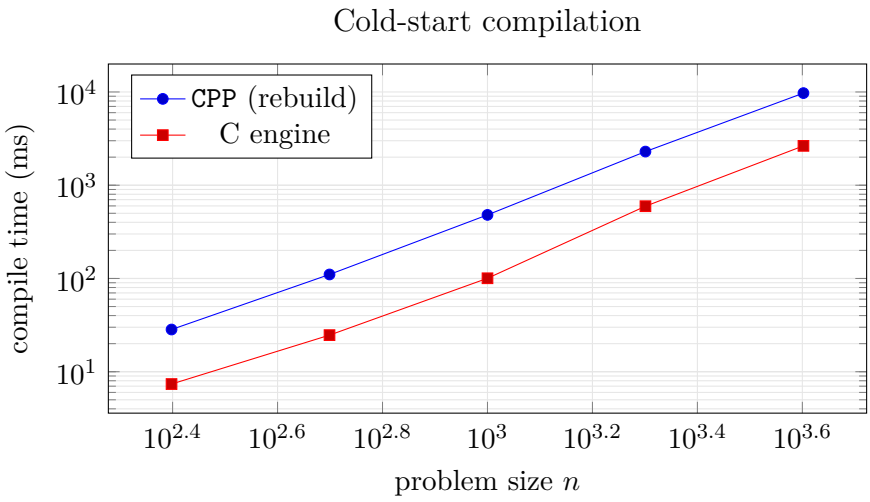



Figure 4.1 Parametric least squares, cold-start: full canonicalization from scratch versus problem size $n$ ($m = 3n$). The C engine is $\approx 4\times$ faster across the whole range; both backends scale as $\sim n^{2.1}$.

### 4.3.2 Factor-covariance portfolio

A Markowitz portfolio whose risk is expressed through a low-rank factor model: $n$ assets load on $m \ll n$ factors with covariance $\Sigma$, plus idiosyncratic diagonal risk $D$. The risk-aversion $\gamma$ and leverage limit $L_{\max}$ are parameters, so re-solving traces the efficient frontier without rebuilding the problem. The size knobs are the number of assets $n$ and factors $m$.

```
import cvxpy as cp
import numpy as np

# mu: expected returns, F: factor loadings (n x m),
# Sigma: factor covariance (m x m), D: idiosyncratic variances (diagonal).
w = cp.Variable(n)                # portfolio weights
f = cp.Variable(m)                # factor exposures
gamma = cp.Parameter(nonneg=True) # risk aversion (reassigned each solve)
Lmax  = cp.Parameter()            # leverage limit (reassigned each solve)

ret  = mu.T @ w
risk = cp.quad_form(f, Sigma, assume_PSD=True) + cp.sum_squares(np.sqrt(D) @ w
    )
objective  = cp.Maximize(ret - gamma * risk)
constraints = [cp.sum(w) == 1, f == F.T @ w, cp.norm(w, 1) <= Lmax]
prob = cp.Problem(objective, constraints)
```

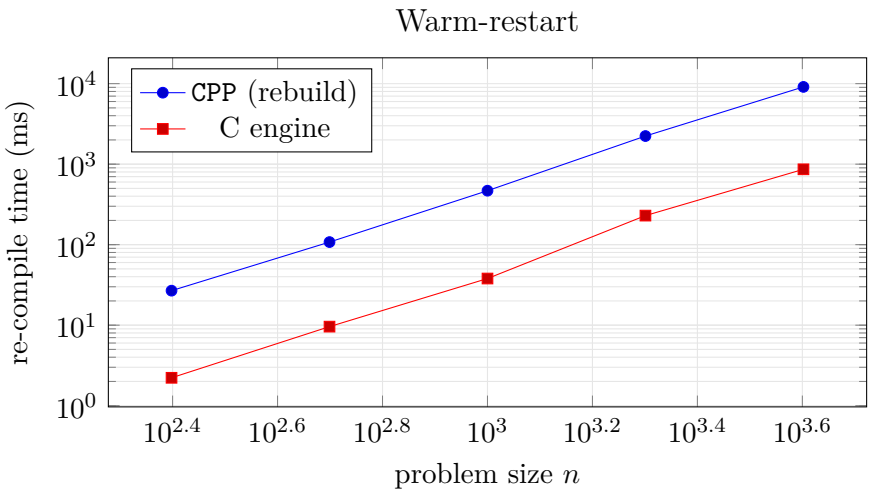



Figure 4.2 Parametric least squares, warm-restart: rebuild after reassigning $A$ and $b$. With no DPP tensor to reuse, `CPP` pays its full cold-start cost again, while the C engine only re-evaluates values on the cached sparsity pattern—a 10–12× speed-up.

We hold the factor count fixed at $m = 100$ and sweep the assets $n \in \{500, 1000, 2000, 4000, 8000\}$. As in the least-squares study, both backends run in the *ignore-DPP* rebuild regime—the engine carries $\gamma, L_{\max}$ natively and re-extracts on a fixed sparsity pattern, while `CPP` substitutes their values and re-canonicalizes. (Here the parameters are scalar, so CPP's DPP cache would in fact be feasible; we use the same rebuild regime as before for a consistent comparison.) Times are medians over three repetitions; peak memory is the process resident-set high-water mark measured in a fresh subprocess.

Cold-start (Figure 4.4) again favours the engine, but less decisively than for least squares and with a narrowing margin: from 0.41× at $n = 500$ to 0.88× at $n = 8000$, as the engine's $\sim n^{1.2}$ cold cost catches CPP's $\sim n^{0.9}$. Warm-restart (Figure 4.5) is where the engine dominates: `CPP` pays its full cold-start cost on every re-solve (144 ms at $n = 8000$), while the engine refills the cached pattern in 10.7 ms—a steady 12–13× advantage. Peak memory (Figure 4.6) is close, the engine 4–16% heavier in resident set; both curves are nearly flat because the low-rank model keeps the operator small. The engine's own internal peak (its `verbose` accounting) grows as $\sim n^{0.9}$, from 6 MB at $n = 500$ to 74 MB at $n = 8000$.

### 4.3.3 Energy-storage scheduling

A simple energy problem: schedule a battery over a horizon of $T$ hours to minimize the cost of grid energy, where the hourly price is the parameter that changes between solves. The state of charge is the running sum of charge minus discharge, and the battery must end

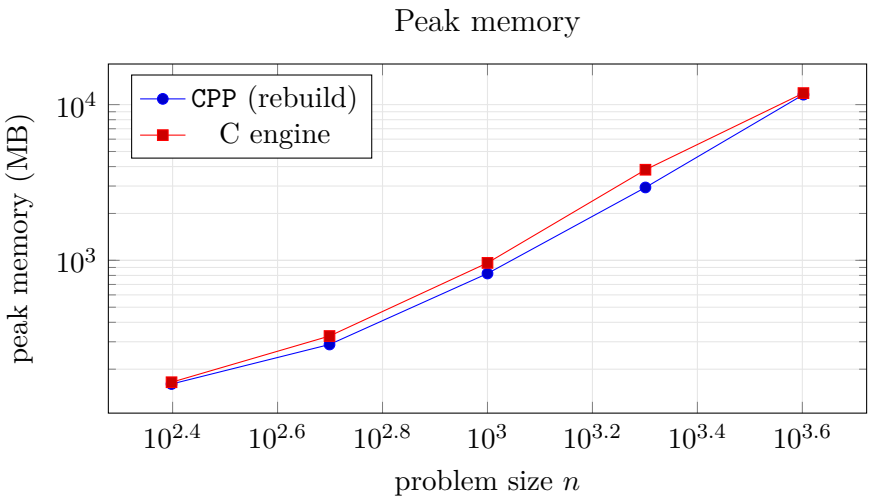



Figure 4.3 Parametric least squares, peak process memory (resident-set high-water mark). The backends are within ∼30% of each other; both grow as $\sim n^{1.6}$, dominated by the shared interpreter and the dense data matrix $A$.

empty. The size knob is the horizon $T$.

```
import cvxpy as cp
import numpy as np

price = cp.Parameter(T)          # hourly price (reassigned each solve)
load  = np.random.rand(T)        # fixed demand profile
Q, Pmax = 50.0, 20.0             # capacity and charge/discharge power limit

charge    = cp.Variable(T, nonneg=True)
discharge = cp.Variable(T, nonneg=True)
soc = cp.cumsum(charge - discharge)     # battery state of charge

constraints = [charge <= Pmax, discharge <= Pmax,
               soc >= 0, soc <= Q, soc[T - 1] == 0]
grid = load + charge - discharge        # net energy drawn from the grid
prob = cp.Problem(cp.Minimize(price @ grid), constraints)
```

The cross-suite results (§4.2) and these scaling curves together delineate the operating regime where the engine pays off: cheap from-scratch builds across the suite, and a favorable size trend on problems whose canonicalization cost the engine avoids re-paying.

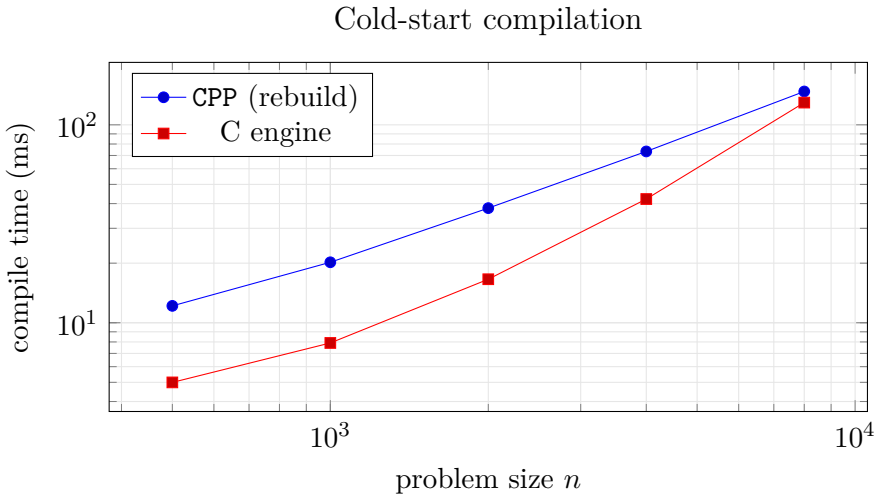



Figure 4.4 Factor-covariance portfolio, cold-start versus number of assets $n$ ($m = 100$ factors). The engine leads at small $n$ but its advantage shrinks with size—it scales as $\sim n^{1.2}$ against CPP's $\sim n^{0.9}$, reaching near-parity by $n = 8000$.

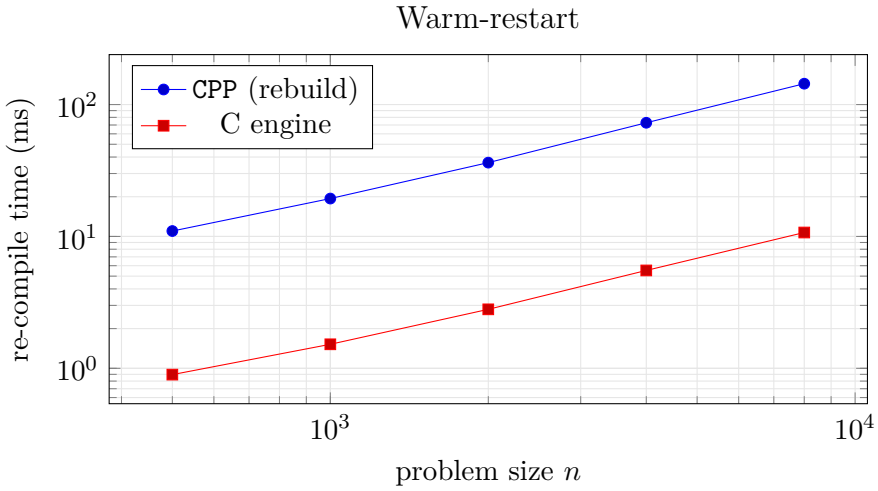



Figure 4.5 Factor-covariance portfolio, warm-restart after reassigning $\gamma$ and $L_{\max}$. `CPP` re-canonicalizes in essentially its cold-start time; the engine re-extracts on the cached pattern, a stable 12–13× speed-up across the range.

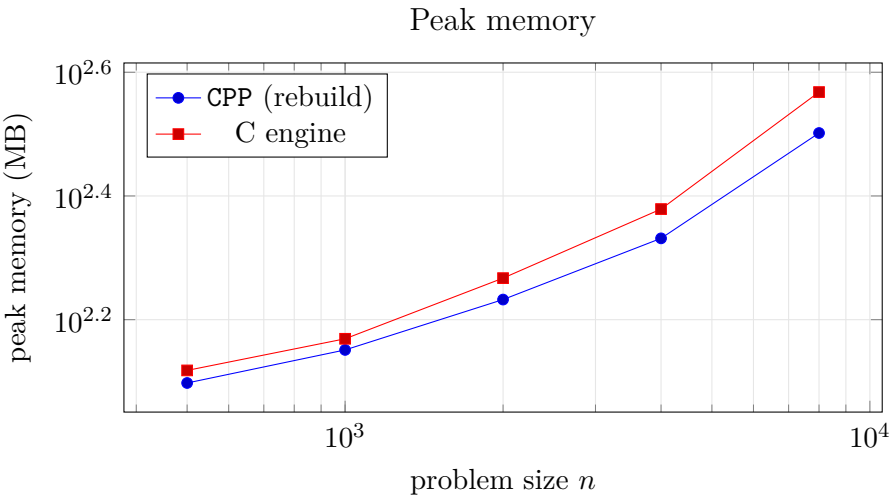



Figure 4.6 Factor-covariance portfolio, peak process memory. The backends are within $\sim$15%; both grow slowly ($\sim n^{0.35}$), reflecting the low-rank factor structure rather than a dense operator.

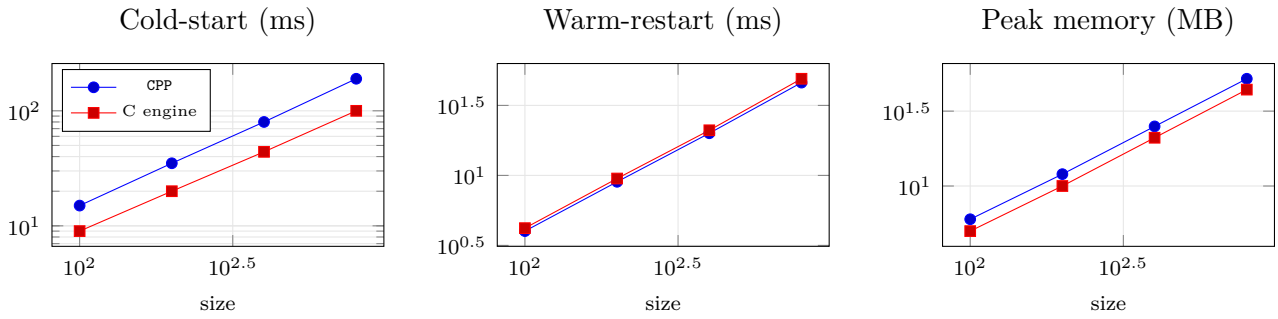



Figure 4.7 Energy-storage scheduling: scaling of cold-start, warm-restart, and peak memory with horizon length $T$ (*placeholder curves; data pending the size-sweep run*).

# CHAPTER 5 CONCLUSION

## 5.1 Summary of Works

## 5.2 Limitations

## 5.3 Future Research

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